

TERRA: an accessible, HPC-integrated pipeline for downstream transposable-element family analysis and CRISPR reagent design

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Abstract

Transposable elements (TEs) make up a large fraction of mammalian genomes and are increasingly recognised as functional contributors to genome regulation in development and disease [1]. Mature tools exist for the *discovery* of TE families from raw assemblies [2], and for quantifying repeat expression [3–5]. What remains under-served is everything that happens once a family has been named: deciding which copies actually matter, reducing a raw annotation to interpretable per-family evidence, and turning that answer into reagents a bench scientist can order. We present **TERRA**, a modular open-source pipeline that takes a TE family name and a reference assembly and returns two concrete reagent deliverables, **family- and locus-specific primers** and **allele-aware CRISPRa/i guide RNAs** scored against the full copy landscape with a coverage-versus-off-target frontier, together with the analytical evidence a bench scientist needs to choose and interpret them. Reaching those deliverables requires resolving a family into sub-groups rather than treating it as one unit, because averaging across a family mixes a minority of expressed, intact copies with a large majority of inert relics. TERRA therefore clusters copies without alignment, separates the *expressed* consensus from the *common* consensus, and layers expression, motif, phylogenetic and structural evidence on each sub-group. Every stage emits figures and machine-readable tables: the cluster embedding, per-cluster expression profiles with differential-expression tests, MAFFT multiple-sequence alignments and per-cluster consensus sequences, JASPAR transcription-factor motif enrichment, and a molecular-clock phylogeny that nominates a master element, alongside the primer and guide tables themselves. We demonstrate the complete path on one prototype family, *L1Md_T* in mm10: 23,639 annotated loci reduce to 5 sequence clusters, 7,490 carry locus-level SQuIRE expression [3], 9388 copies retain ORF integrity, and the pipeline nominates a top master-element candidate (chrX:40304675) and 400 candidate guides of which 17 are non-dominated on the coverage/off-target frontier. The same run executes unchanged on a laptop or on LSF/Slurm clusters through a Nextflow DSL2 layer [6, 7]. TERRA is open source.

1 Introduction

Interspersed repeats derived from TEs account for roughly half of the human genome [8] and a comparable or larger share of many other eukaryotic genomes [1], where they shape genome organisation, supply regulatory sequence, and influence host gene expression in development and disease [9, 10]. The natural unit of analysis is the *family*, which groups copies descended from a common ancestral element.

The bioinformatics of TEs has historically concentrated on two upstream problems: building libraries of family models *de novo* from an assembly, and annotating a genome against them. Tools such as RepeatModeler2 [2] and the RepeatMasker/Dfam ecosystem [11] address

both well. A second, well-populated group quantifies repeat expression, including SQuIRE [3], TEtranscripts/TElocal [4], Telescope [5], scTE [12] and ERVmap [13].

What is comparatively under-served is the workflow that begins once a family has been named and quantified, and it is the workflow that determines whether a biologist can act. Two problems sit at its centre.

The first is determining which copies truly matter. A named family is not a homogeneous object: it bundles a small number of expressed and intact insertions together with a much larger number of truncated and inert relics. Analyses that treat the family as one unit average across all of them, so a real signal carried by a minority of loci is diluted by the majority that carry none. Resolving the family into sub-groups and then asking which sub-groups are expressed is therefore not a convenience; it is what makes the downstream result interpretable, and it produces the per-family evidence a biologist reads directly: the cluster embedding, per-cluster expression, multiple-sequence alignments and consensus sequences, and motif enrichment.

The second is that the end point of most TE projects is experimental, which requires sequence-level reagents, and repetitiveness is exactly what makes them hard to design. A primer pair must be specific enough to report on the intended copies, and a CRISPRa/i guide will, by construction, match many copies at once, so its value depends on an explicit trade-off between how much of the family it covers and how many off-target sites it hits. TERRA treats primers and guides as first-class outputs rather than an afterthought, and reports the guide trade-off as a Pareto frontier rather than a single score. Both the reagents and the analytical evidence behind them, namely the alignments, consensus sequences, expression profiles, motif enrichment and phylogeny, are returned as figures and tables rather than left for the user to assemble.

TERRA applies the philosophy of integrated analysis platforms [6, 7, 14, 15], namely hide the infrastructure, keep every step reproducible, and run anywhere, to this specific problem. It is not a TE-discovery tool and does not compete with RepeatModeler2 [2]. It begins with a defined family and assembly, and automates the characterisation and reagent design that follow, with first-class support for the LSF and Slurm schedulers that dominate academic HPC.

Figure 1 summarises the path this paper follows end to end on a single prototype family.

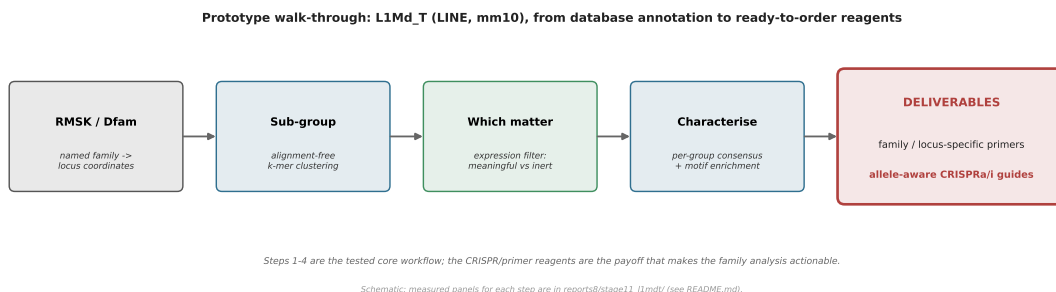


Figure 1: The TERRA path, from database annotation to ordered reagents, illustrated for the prototype family used throughout this paper. RepeatMasker/Dfam loci are sub-grouped without alignment, filtered by locus-level expression to separate copies that matter from inert relics, characterised by consensus and motif content, and converted into the two deliverables: family/locus-specific primers and allele-aware CRISPRa/i guides. This panel is a schematic of the workflow; every measured result it refers to is reported in Section 2.

2 Results: L1Md_T from annotation to reagents

We use one family, *L1Md_T*, as a prototype throughout. *L1Md_T* is a young, active mouse LINE-1 subfamily, which makes it a demanding test: it is large, it contains both intact and heavily truncated copies, and it is expressed in early development, so every stage of the workflow is exercised. All results below come from a single `mm10` run with locus-level expression quantified by SQuIRE [3] across six stages of early mouse development (oocyte, pronucleus, 2-cell, 4-cell, 8-cell, morula). Results for a SINE (*B1_Mus2*) and an LTR (*IAPLTR1_Mm*) family, which establish that the same workflow generalises across TE classes, are summarised in Section 7 and provided in full as Supplementary Files.

2.1 Starting point: the annotated family

TERRA takes the family name and assembly and retrieves loci and per-locus divergence from the RepeatMasker track [11], then extracts sequence from the reference. For *L1Md_T* in `mm10` this yields 23,639 annotated loci. This is the number a biologist starts with, and on its own it is not actionable: it says nothing about which of those 23,639 insertions are intact or worth targeting.

2.2 Resolving copies rather than lumping them

Clustering is alignment-free. Each locus sequence is encoded as a k -mer count vector (default $k=6$, giving a $4^6 = 4096$ -dimensional sparse vector built with scikit-learn [16]), reduced to 50 components by truncated SVD, embedded in two dimensions with UMAP [17], and partitioned with HDBSCAN [18]. This avoids a family-wide multiple alignment and scales to tens of thousands of loci. For *L1Md_T* the parameter search (Section 4) converged to a divisor of $N/5$ with UMAP `n_neighbors = 50`, $k = 5$, minimum cluster size 2000 and `min_samples = 25`, partitioning the family into 5 clusters with 7.2% of loci left in HDBSCAN’s noise class.

These parameters were not taken on faith. The pipeline’s own heuristic fixes the minimum cluster size at $\lfloor N/5 \rfloor$, which for *L1Md_T* demands 4,727 loci per cluster: only 2 groups can satisfy that threshold, so everything else is forced into the noise class and 53.8% of the family ends up unassigned. That is a property of the threshold rather than of the family. We therefore searched k , `n_neighbors`, `min_cluster_size` and `min_samples` (288 configurations, `run_cluster_search.py`) for a partition with a plausible number of sub-groups at minimal noise; 44 configurations recovered 5 to 6 clusters, and the selected one reduces the unassigned fraction from 53.8% to 7.2% (Figure 6). The noise class that remains is retained and reported as a labelled group, and is excluded from per-cluster consensus building (Section 4), so unassigned loci cannot contaminate a consensus that reagents are designed against.

The reason to do this is shown in Figure 2. Treating the family as a single unit averages across three quite different populations: a minority of expressed, intact copies; a set of low-function copies; and a large majority of inert relics. Any family-level mean is dominated by the third group, so a real effect carried by the first is lost. Once copies are resolved into sub-groups and intersected with expression, the populations can be handled separately, and the consensus built from expressed copies (the *expressed* consensus) is the sequence that primers and guides should target, which is not in general the same as the consensus built from all annotated copies.

Of the 23,639 *L1Md_T* loci, 7,490 carry locus-level expression entries in the SQuIRE table, that is, roughly a third of annotated copies contribute any measured signal. Expression is reported per locus as SQuIRE counts and is summarised per cluster; units and the exact counting convention

Resolving copies, not lumping them — a core conceptual choice

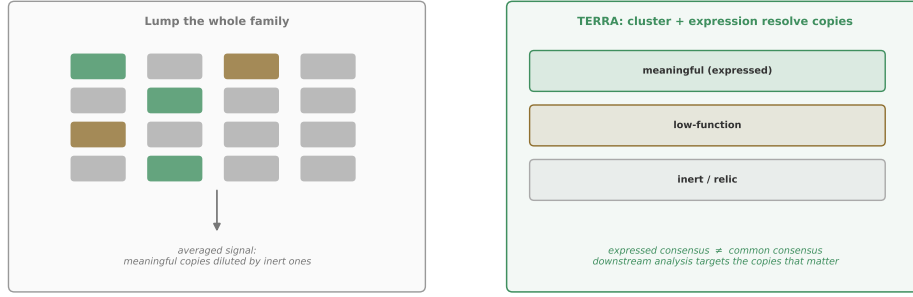


Figure 2: Resolving copies instead of lumping them. Left: a family-level summary averages over meaningful, low-function and inert loci, so signal carried by a minority of copies is diluted. Right: TERRA partitions copies by sequence and intersects them with locus-level expression, separating the populations and distinguishing the expressed consensus from the common consensus. This panel is a schematic of the concept; the measured partition for *L1Md_T* is in Figure 3.

are stated in Section 4. Pairwise Wilcoxon rank-sum tests between clusters, corrected with the Benjamini–Hochberg procedure, find 54 of 60 cluster-pair by stage comparisons significant at $q < 0.05$, across all 6 developmental stages and every pair of the 5 clusters. Expression is not spread evenly over the clusters: it concentrates in a minority of them and peaks at the 8-cell stage, while the remainder are close to silent (Figure 3b). This is the separation that a family-level average would remove. Figure 3 shows the measured partition, the per-cluster expression profile, and the divergence distribution from this run.

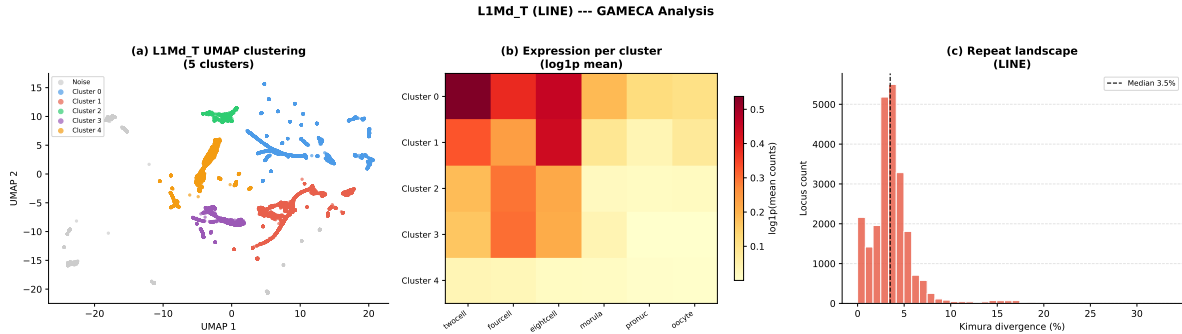


Figure 3: *L1Md_T* (mm10) analysed end to end in a single TERRA run. (a) UMAP embedding of k -mer-encoded loci coloured by HDBSCAN cluster. (b) Per-cluster mean SQUIRE expression ($\log_{1p}$ mean) across the six developmental stages, computed from the 7,490 loci carrying expression entries. (c) CpG-corrected Kimura divergence histogram, median 3.5%.

2.3 Age, intactness, and the master element

Copies are aligned with MAFFT [21], a majority-rule consensus is built, and each copy’s Kimura two-parameter divergence [22] from that consensus is converted to a molecular-clock age. ORF integrity separates intact from degraded copies, and a master-element score combining intactness and divergence nominates the youngest, most intact copies, which are the ones most likely to still be capable of mobilisation. For *L1Md_T*, 9388 of 23,639 copies retain ORF integrity, the median

copy age is 5.41 Myr, and the top master-element candidate is chrX:40304675 at 7.23% divergence from the consensus (Figures 8 and 4).

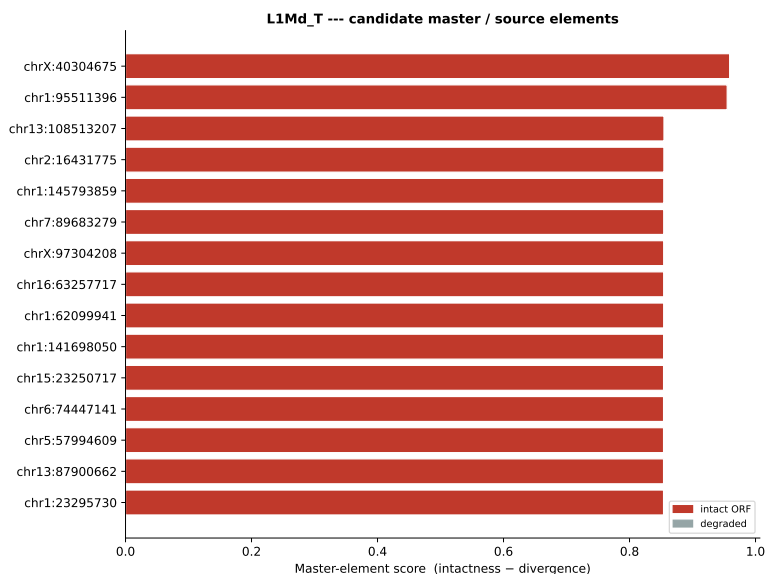


Figure 4: Master-element nomination for *L1Md_T*. Copies are scored on ORF integrity against divergence from the consensus; 9388 copies are intact and the top-ranked candidate is chrX:40304675.

2.4 Deliverable 1: allele-aware CRISPRa/i guides

Designing CRISPR reagents against a repetitive family is the problem that most tools leave to the user, and it is the capability that makes this pipeline useful at the bench rather than merely descriptive. A guide targeting a TE family does not have a single on-target site: it matches some number of family copies and some number of sites elsewhere. The design question is therefore explicitly bi-objective, namely maximise on-family coverage while minimising off-target hits, and it cannot be reduced to one score without hiding the trade-off from the user.

TERRA enumerates PAM-aware candidate guides from the consensus and from per-copy sequence, then scores every candidate against the *full* copy landscape (and an optional background FASTA) using seed-anchored, mismatch-tolerant matching, and reports the coverage-versus-off-target Pareto frontier. For *L1Md_T*, 400 candidate guides were scored against all 23,639 copies, of which 17 are non-dominated and therefore represent the genuine design choices (Figure 5); the best-scoring guide on that frontier is AGGGAGGGAAGGAGGGAGGG. A user selecting from that frontier is choosing a documented position on the coverage/specificity trade-off rather than trusting an opaque ranking. The same output supports CRISPRa and CRISPRi, since both depend on the guide-to-copy mapping rather than on the effector. Figure 9 summarises the design logic.

2.5 Deliverable 2: family- and locus-specific primers

Primers are proposed from frequent *k*-mers in the (expressed) consensus and each candidate is checked for specificity genome-wide across all chromosomes. Because the specificity check is the most I/O-intensive step in the pipeline, it motivates the genome-cache engineering described in Section 5. Primer candidates are emitted as a table with their genome-wide match counts, and are

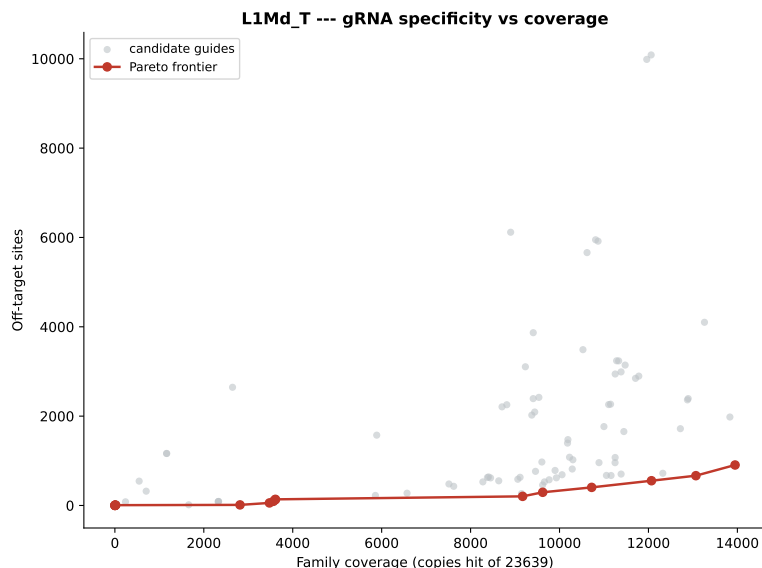


Figure 5: Measured coverage-versus-off-target frontier for *L1Md_T*. 400 candidate guides scored against 23,639 copies; 17 guides are non-dominated. Each frontier point is a defensible design choice at a stated coverage and off-target count.

cross-referenced against the per-locus expression so that a user can select primers that report on the copies that are actually transcribed rather than on the family as annotated.

2.6 Additional copy-resolved evidence

Three further modules were exercised on *L1Md_T* and are reported here because they bear directly on reagent design. 3' transduction detection groups copies sharing downstream-tail k -mers absent from the consensus, alongside poly-A signal detection, and resolves 30 candidate lineages with 8227 copies carrying an AATAAA signal (Figure 10). Antisense/bidirectional promoter scanning of each 5' region on both strands flags 13246 of 23,639 copies as bidirectional (Figure 11), which is relevant because a CRISPRa guide placed in a bidirectional promoter may drive transcription in both directions. A further set of modules (CTCF/TAD proximity, epigenetic overlay, orthologous-insertion calling, multi-assembly liftover, ColabFold structure prediction, LTR structural classification, and automatic subfamily resolution) place the same resolved copies in regulatory, evolutionary and structural context, and are described in Section 6.

3 How TERRA relates to existing tools

TERRA's neighbours fall into three groups (Table 1), and it consumes the output of the first two rather than competing with them.

Discovery and annotation. RepeatModeler2 performs *de novo* discovery and emits consensus models [2], which the RepeatMasker/Dfam ecosystem uses to annotate a genome [11]. Both produce inputs to family characterisation rather than instances of it, and TERRA consumes RepeatMasker/Dfam locus coordinates as its starting point.

Repeat expression quantification. A well-established group of tools assigns reads to repeats: SQuIRE reports locus-specific repeat expression [3]; TETranscripts and TELocal handle multi-

mapping reads at family and locus level [4]; Telescope reassigns reads across the retrotranscriptome [5]; scTE extends TE quantification to single-cell data [12]; ERVmap targets human endogenous retroviruses [13]; and TE-Seq runs an end-to-end gene and TE RNA-Seq pipeline with TE-aware annotation [23]. TERRA consumes per-locus counts from any of these (SQuIRE in this paper) and does not perform quantification itself. These tools answer “how much is expressed”; TERRA addresses what follows, namely which copies to group together, what their consensus is, and what reagent will target them.

General platforms. Galaxy [14], GenePattern [15] and nf-core/Nextflow [6, 7] are portable, reproducible substrates, but none ships a ready-made downstream TE-family workflow; the user must locate, wrap and wire each stage. TERRA builds on this substrate rather than competing with it: it ships a Nextflow DSL2 implementation that follows nf-core conventions and is importable as a subworkflow.

The comparison in Table 1 is one of *scope and access*, compiled from each tool’s documented capabilities. No cross-tool runtime is claimed, and the runtime figures in Section 5 concern TERRA’s own engineering only.

Tool	TE-family focus	Copy-resolved character.	Reagent design	Graphical interface	Built-in LSF/Slurm	Reprod. environ.
TERRA (this work)	✓	✓	✓	✓	✓	✓
RepeatModeler2 [2]	✓	×	×	×	×	○
RepeatMasker/Dfam [11]	✓	×	×	×	×	○
SQuIRE [3]	✓	○	×	×	×	○
TEtranscripts/TElocal [4]	✓	○	×	×	×	○
Telescope [5]	✓	○	×	×	×	○
scTE [12]	✓	○	×	×	×	○
ERVmap [13]	✓	○	×	×	×	○
TE-Seq [23]	✓	○	×	×	○	○
Galaxy [14]	×	○	×	✓	○	✓
GenePattern [15]	×	○	×	✓	○	✓
nf-core/Nextflow [7]	×	○	×	×	✓	✓

Table 1: Capability comparison on documented features. ✓ built-in; ○ partial, or possible but the user must assemble or configure it; × not provided. *Copy-resolved characterisation*: partitioning a named family into sub-groups and building per-group consensus, as distinct from quantifying expression per locus. *Reagent design*: primers and/or allele-aware CRISPR guides emitted for the family. Expression quantifiers are marked ○ for copy-resolved characterisation because they report per-locus or per-family counts without grouping copies or building consensus. No cross-tool runtime is claimed.

4 Validation and reporting conventions

Units. Expression is consumed as per-locus counts from an external quantifier (SQuIRE [3] throughout); TERRA does not requantify. Per-cluster values are means of $\log_{1p}$ per-locus counts over the loci in that cluster carrying an entry; loci with no entry are excluded rather than treated as zero, which is why 7,490 of 23,639 loci enter the analysis. Motif results are counts of JASPAR matches [24] per locus intersected with BEDTools [25]; enrichment is a per-cluster Fisher exact test on loci-carrying-a-match, so a locus with several matches for one motif contributes once.

The noise class. HDBSCAN labels loci it cannot confidently place as -1 . These are retained and reported as a labelled group but are excluded from per-cluster consensus building, so a locus never contributes to a consensus it was not assigned to. The adopted configuration leaves 7.2% of loci unassigned.

The partition is a stratification of copy state, not a subfamily assignment. Three measurements bound what the clusters can be claimed to be, and we report them including the negative results (Appendix A). *Parameter dependence:* across 9 configurations of k and `n_neighbors` the recovered cluster count ranged from 2 to 6 with a median ARI of 0.40 against the adopted partition, an improvement on the pipeline’s superseded default (median ARI 0.21, which admitted only 2 clusters and left 53.8% unassigned) but short of a subfamily claim. *Reproducibility:* clustering is deterministic within an environment (re-running returns identical labels, ARI 1.000), but UMAP runs with `n_jobs=-1`, so cluster counts can shift on hardware with a different core count; every figure here was produced from the stored labels and embedding rather than re-derived. *Consensus cross-check:* a consensus was built per cluster and every sampled copy projected onto every consensus with `mafft -addfragments -keeplength`; only 36.1% of copies are nearest their own cluster’s consensus (chance 20.0%), and the median copy is marginally *closer* to another cluster’s consensus (0.035 versus 0.030). The clusters therefore do not correspond to distinct consensus sequences and we do not call them subfamilies anywhere in this paper. What they do capture, sharply, is degradation state: ORF integrity runs from 91.2% intact in the most intact cluster to 0.1% in the least, and expression concentrates in the intact-rich clusters. That is the axis the workflow needs, since separating expressed, intact copies from inert relics is what makes the downstream reagents meaningful.

Scope. The core workflow together with the phylogenetics, gRNA, transduction and antisense modules is the source of every number reported above; the further analyses of Section 6 extend the same resolved family into regulatory, evolutionary and structural context.

5 Design and implementation

TERRA is organised in three layers (Figure 14). A desktop application built with Tauri and a React front end gives point-and-click access and performs no analysis itself; it launches a bundled Python sidecar and communicates over newline-delimited JSON on standard streams, so long-running jobs stream progress and can be cancelled. The analysis backend is a driver (`query.py`) that orchestrates the core stages, one module per stage, a family of standalone `run_*.py` modules, and an SSH client. In local mode everything runs on the user’s machine, fetching sequence from the UCSC API [26] when no local FASTA is available. In HPC mode the code is staged to a login node, submitted as a batch job, and the results retrieved. Every external resource (RMSK/Dfam loci, UCSC sequence, JASPAR [24], mygene.info [27]) is reached through a thin adapter, so adding an assembly or swapping a source is a localised change.

The pipeline is a directed sequence of stages (Table 2). Each stage reads the previous stage’s tabular output and writes a checkpoint, so a run can be resumed or a single stage re-executed. Figure 15 gives the end-to-end view.

Engineering for the bottleneck. The dominant cost in whole-family primer design is reading the reference genome, since a naive implementation re-reads the multi-gigabyte FASTA for every candidate search. TERRA parses the reference once into an in-memory genome cache, reused for

Stage	Function	Key methods / tools
Prep	Loci, sequences, divergence	RMSK/Dfam [11]; FASTA or UCSC [26]; Kimura milliDiv
Clustering	Sub-group the copies	k -mers $\rightarrow$ SVD $\rightarrow$ UMAP [17] $\rightarrow$ HDBSCAN [18]
Alignment	Consensus per group	MAFFT [21] + CIALign [28]
Motif	Enriched TFBS	JASPAR [24] $\cap$ loci (BEDTools [25]) + Fisher
GO	Gene/ontology context	mygene.info [27]
Expression	Per-group profiles + DE	per-locus counts (SQuIRE [3]); Wilcoxon + BH
Primers	Family/locus primers	k -mer candidates + genome-wide specificity
gRNA	Allele-aware CRISPRa/i	PAM-aware candidates + seed-anchored off-target scoring

Table 2: The TERRA core stages. Each stage consumes a tabular checkpoint and emits one, allowing partial re-runs. The gRNA and primer stages produce the deliverables of Section 2.4.

sequence extraction and all primer searches, and persists it so later runs skip parsing. A small number of batch operations (GC content, length tabulation, k -mer matrix construction) dominate the per-sequence arithmetic and have Cython [29] implementations with a pure-Python fallback.

Orchestration. The driver can run the copy-resolved modules as a sequential subprocess loop, or delegate them to a Nextflow DSL2 layer [6] that runs the same unmodified Python modules as parallel, individually-resourced, resumable tasks (Figure 16). No scientific code is duplicated between the two.

6 Additional analysis modules

Around the core walk-through TERRA ships a set of modules that extend the same resolved family into its regulatory, evolutionary and structural context. Each runs as an independent, individually-resourced task, consumes the same per-copy tables as the core stages, and emits its own figures and machine-readable values; every run also writes a provenance manifest (git SHA, platform, tool versions) alongside the checkpoints that let it resume (Figure 17). They fall into three groups.

Regulatory context. CTCF-site and TAD-boundary proximity flags copies sitting at or near architectural boundaries, and an epigenetic and regulatory-track overlay intersects copies with user-supplied signal tracks. Together these separate silenced from active insertions and so complement the expression stratification of Section 2.2 with orthogonal regulatory evidence.

Evolutionary context. Orthologous-insertion calling and multi-assembly liftover across hg38, T2T-CHM13, mm10 and mm39 place each copy in other assemblies and species, distinguishing shared ancestral insertions from lineage-specific ones. Automatic subfamily resolution collapses the per-cluster consensus into a dendrogram, giving a coarse view of relatedness above the level of the individual copy.

Structural context. A pure-Python LTR structural classifier labels each element full-length, solo-LTR, internal-only or truncated, and ColabFold structure prediction folds the consensus ORFs, extending the characterisation from sequence to protein structure and, for the LTR case, to the intact-element architecture that determines mobilisation competence.

Each of these runs on any family the core workflow accepts and writes the same figure-and-table output as the core stages, so they slot into the same per-family report.

7 Generality across TE classes

To establish that the workflow is not specific to a LINE, we applied the identical pipeline to a SINE and an LTR family in *mm10* with the same six-stage SQuIRE dataset. *B1_Mus2* (SINE) has 70,685 loci with 21,407 carrying expression and a median Kimura divergence of 7.5%. *IAPLTR1_Mm* (LTR) has 1,485 loci with 1,426 carrying expression and a median divergence of 1.6%. The divergence medians order the three families by evolutionary age, with *IAPLTR1_Mm* youngest and most homogeneous, *L1Md_T* intermediate at 3.5%, and *B1_Mus2* oldest and most diverged, which is consistent with their known retrotransposition histories [1]. Full per-family reports for all three families, including the clustering, expression, alignment, consensus, motif, primer and guide outputs, are provided as Supplementary Files.

8 Discussion

TERRA fills a specific gap: the accessible, reproducible, cluster-ready path from a named TE family to reagents that can be ordered. It complements rather than replaces discovery tools such as RepeatModeler2 [2], expression quantifiers such as SQuIRE [3] and Tetrascripts [4], and general platforms such as Galaxy [14] and nf-core [7].

Its contributions are threefold. First, a conceptual one: resolving a family into sub-groups and intersecting them with locus-level expression, rather than summarising the family as a unit, is what allows a signal carried by a minority of copies to survive the analysis, and it motivates the distinction between the expressed and the common consensus (Section 2.2). Second, a practical one: treating primers and allele-aware CRISPRa/i guides as the deliverable, and reporting the guide coverage/off-target trade-off explicitly as a frontier rather than collapsing it into one score (Section 2.4). Third, an engineering one: the in-memory genome cache, the compiled hot paths, and automatic LSF/Slurm support make whole-family analysis practical without rebuilding the workflow when moving from a laptop to a cluster.

Several limitations bound the system. Expression analysis requires user-supplied per-locus counts, so quantification quality and its multi-mapping policy are inherited from the upstream tool. Family resolution and sequence depend on external services (RMSK/Dfam, UCSC, JASPAR, mygene.info), so a run is only as reproducible as the database releases it draws on, and supported assemblies are currently the common human and mouse builds. The *k*-mer clustering, while fast and alignment-free, does not yield stable sub-family boundaries at the default minimum cluster size: agreement between parameter settings is close to chance (median ARI 0.40, Section 4), so we report the partition as a device for separating expressed from inert copies rather than as a subfamily assignment. Resolving this requires the parameter search described in Section 4 followed by cross-validation against the consensus alignments. The guide frontier likewise reports sequence-level predictions that require experimental confirmation.

Natural extensions follow from the architecture: per-locus epigenomic context (WGBS, ATAC-seq) to separate silenced from derepressed copies; correlating per-cluster TE activity with nearby gene expression to nominate regulatory TEs; somatic-insertion calling from user BAMs; broader assembly support; and negative-binomial modelling of count overdispersion using established frameworks [30, 31].

Code availability

TERRA is open source at <https://github.com/ammol-dash/te-scraper-and-analysis>. Figures are generated by the pipeline scripts in that repository, which run the analysis locally or on an HPC cluster and emit the figures and measured values cited here.

A Supplementary figures

The figures below support the results and validation reported in the main text. They are placed here to keep the main narrative on the prototype walk-through and the reagent deliverables; each is referenced from the section it belongs to.

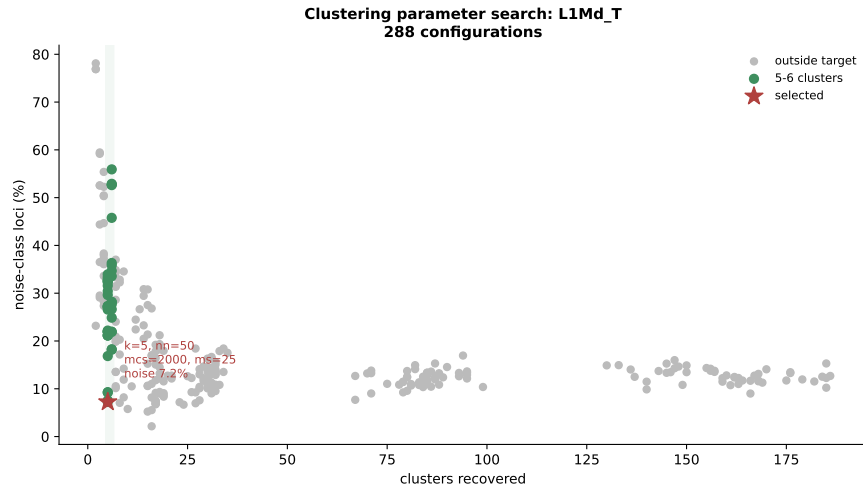


Figure 6: Clustering parameter search for *L1Md_T*. Each point is one of 288 configurations of k , $n_neighbors$, $min_cluster_size$ and $min_samples$, plotted as the number of clusters recovered against the fraction of loci left unassigned. The pipeline’s default minimum cluster size ($\lfloor N/5 \rfloor = 4,727$) admits only 2 clusters and leaves 53.8% of the family unassigned; the selected configuration (star) recovers 5 clusters with 7.2% unassigned.

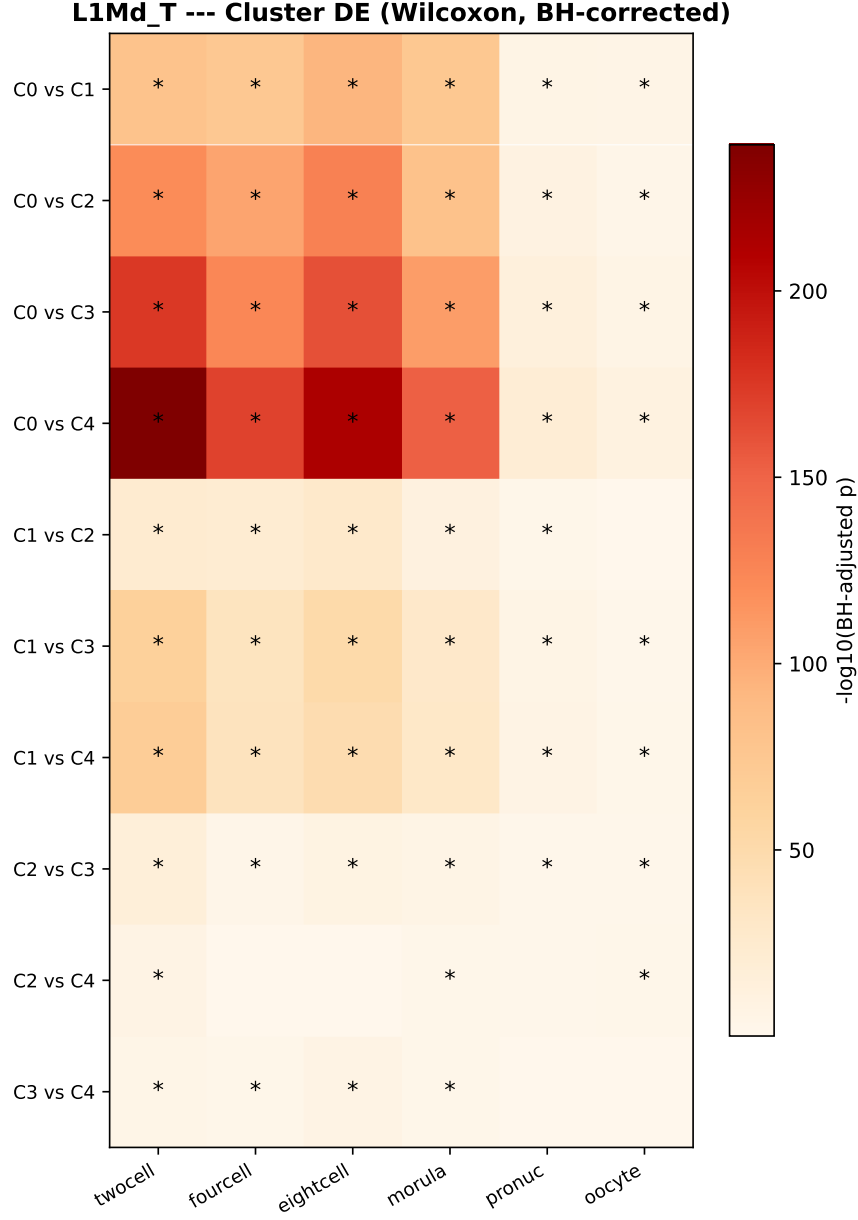


Figure 7: Differential expression between *L1Md_T* clusters. Heatmap of $-\log_{10}(q)$ from pairwise Wilcoxon rank-sum tests with Benjamini–Hochberg correction; 54 of 60 cluster-pair by stage comparisons reach $q < 0.05$. Cluster-resolved testing recovers stage-specific differences that a family-level summary averages away (Section 2.2).

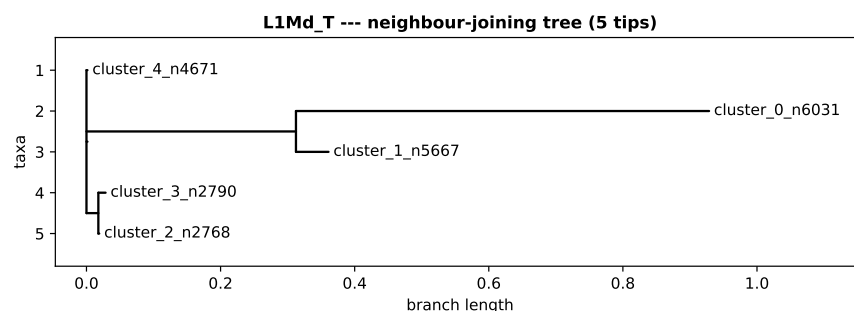


Figure 8: Subfamily phylogeny for *L1Md_T*. Neighbour-joining tree relating cluster consensus and representative copies, built from the MAFFT alignment.

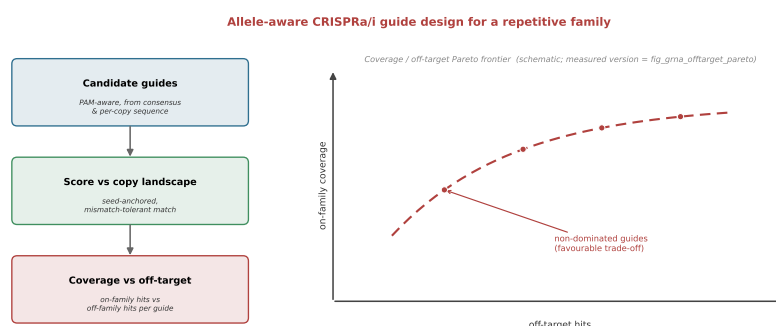


Figure 9: Allele-aware CRISPRa/i guide design for a repetitive family. Candidate guides are enumerated PAM-aware from the consensus and per-copy sequence, scored against the full copy landscape with seed-anchored, mismatch-tolerant matching, and reported as a coverage-versus-off-target trade-off. This panel is a schematic of the logic; the measured frontier for *L1Md_T* is Figure 5.

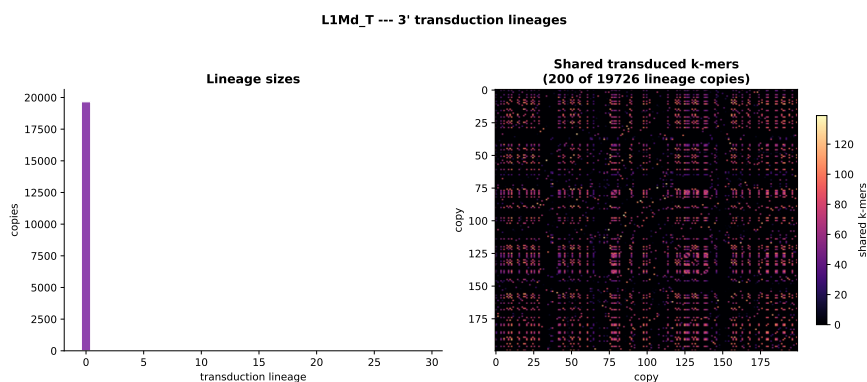


Figure 10: 3' transduction lineages for *L1Md_T*: 30 candidate lineages grouped by shared downstream-tail *k*-mers absent from the family consensus; 8227 copies carry an AATAAA poly-A signal.

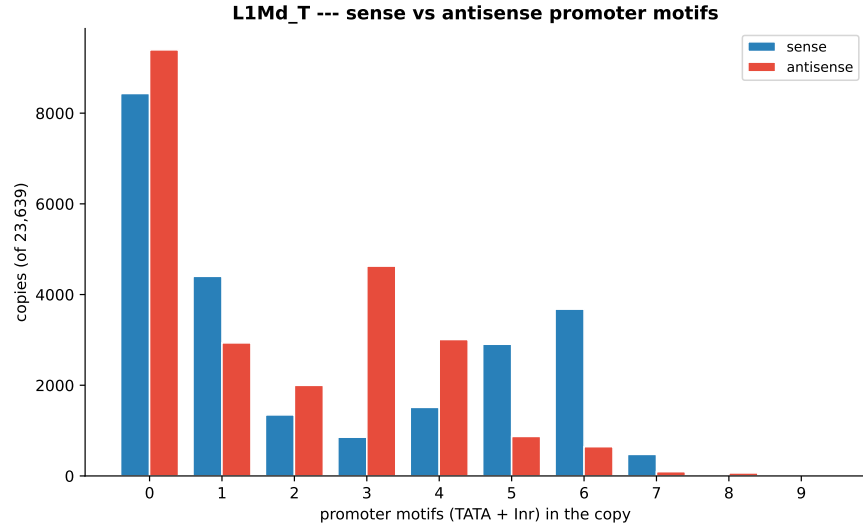


Figure 11: Antisense/bidirectional promoter content for *L1Md_T*: core promoter elements scanned on both strands of each 5' region; 13246 of 23,639 copies are flagged bidirectional.

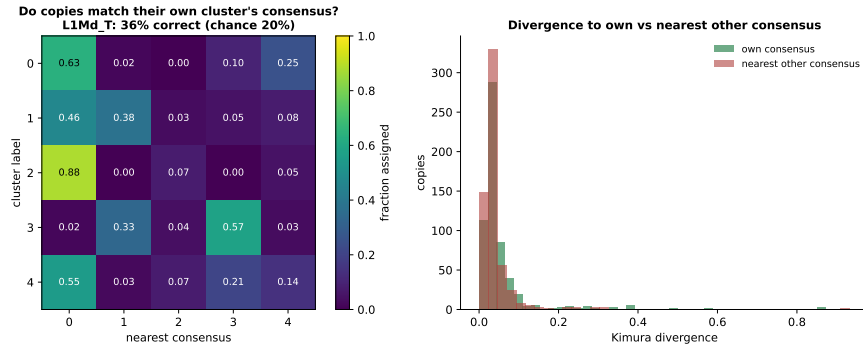


Figure 12: Cross-check of the clusters against the consensus alignments for *L1Md_T*. **Left:** fraction of each cluster's copies whose nearest consensus is each cluster's, after projecting every copy onto every consensus with `mafft -addfragments -keeplength`. A subfamily assignment would put the mass on the diagonal; the observed 36.1% (chance 20.0%) does not. **Right:** Kimura divergence of each copy to its own cluster's consensus versus to the nearest other consensus; the distributions overlap, with the median copy marginally closer to another cluster's consensus.

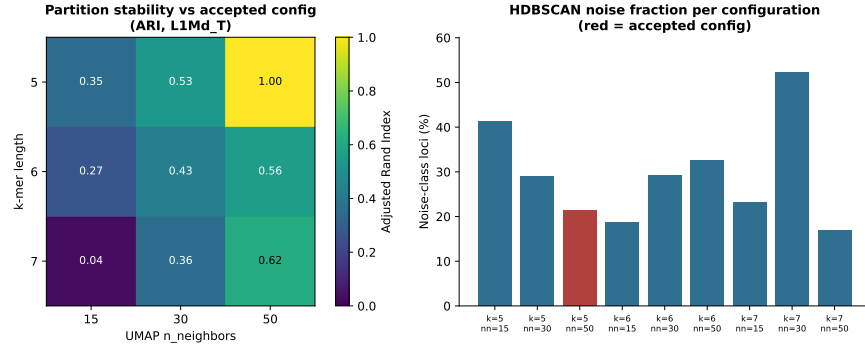


Figure 13: Clustering and noise-class validation for *L1Md_T*. **Left:** Adjusted Rand Index of each k and $n_neighbors$ configuration against the configuration the automatic search accepted. Values near 1 would indicate the partition is independent of the parameter choice; the observed median of 0.40 indicates it is not, which is why these sub-family boundaries are not interpreted as biological. **Right:** HDBSCAN noise-class fraction per configuration, with the accepted configuration highlighted. Noise-class loci are retained as a labelled group and excluded from consensus building.

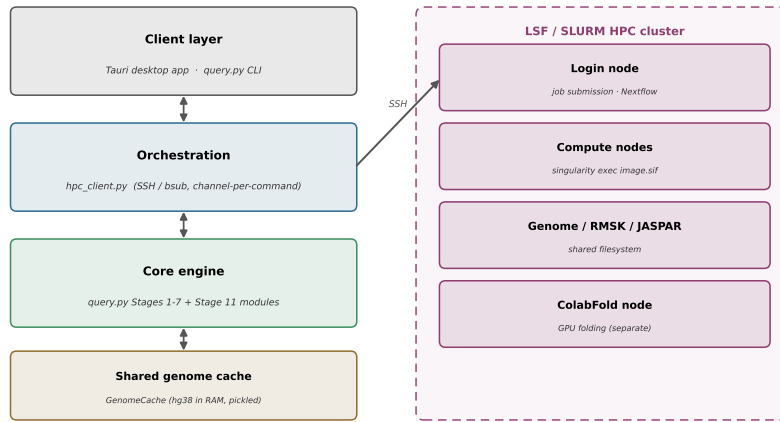


Figure 14: TERRA's three-layer architecture. A Tauri/React desktop shell drives a Python analysis sidecar over a streaming NDJSON protocol; the backend runs locally or stages and submits jobs to an LSF/Slurm cluster. Coordinate, sequence, motif and annotation data are pulled from RMSK/Dfam, UCSC, JASPAR and mygene.info through uniform adapters.

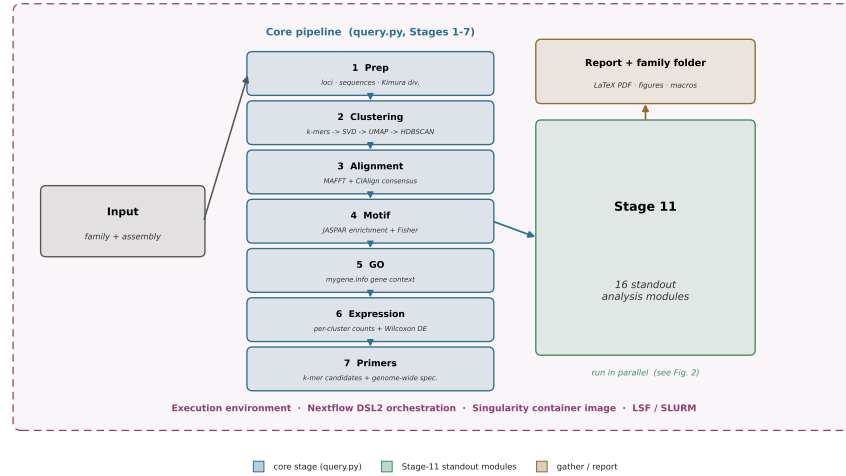


Figure 15: TERRA pipeline flowchart. A family/assembly request flows through the core stages and into the copy-resolved modules, which run in parallel; outputs are gathered into a per-family folder and report. The run executes inside the Nextflow DSL2 and Singularity environment on an LSF or Slurm scheduler.

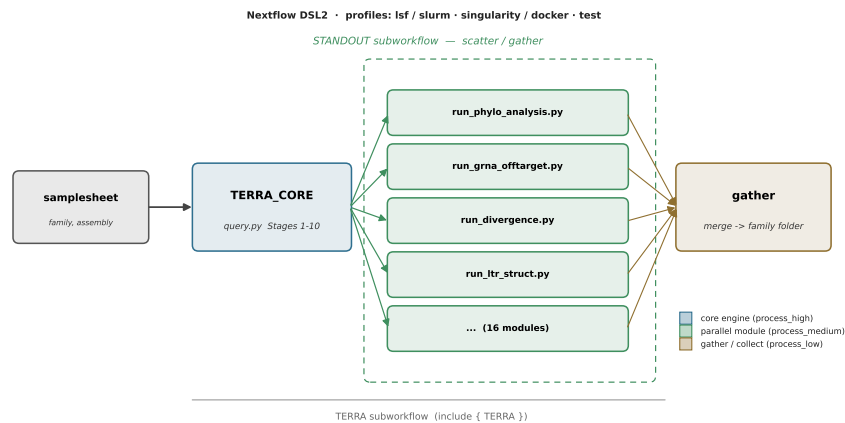


Figure 16: The Nextflow DSL2 execution graph: the core engine feeds a scatter of independent copy-resolved module tasks, gathered into a single per-family output. Profiles select LSF or Slurm and Singularity or Docker.

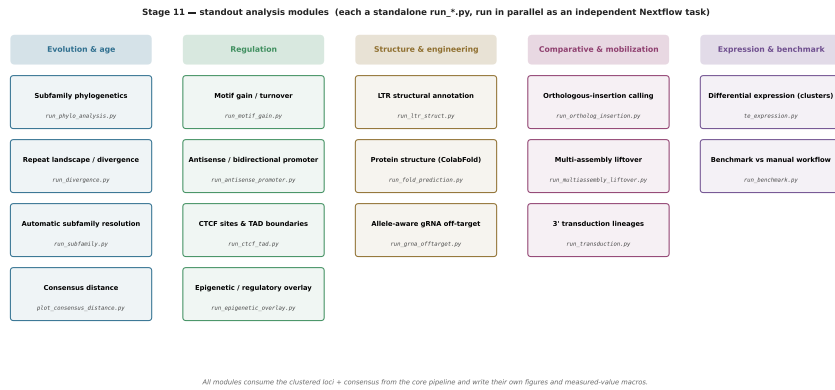


Figure 17: The copy-resolved analysis modules grouped by biological theme. Each card names the standalone `run_*.py` script. The phylogenetics, gRNA, transduction and antisense modules are reported as results in Section 2; the remainder are the additional analysis modules of Section 6.

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